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A Comparative Evaluation of Nanoparticle-Based Drilling Fluids for HPHT Wells: A Meta-Analytical Approach

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Abstract

This study presents a comprehensive meta-analysis of nanoparticle-enhanced drilling fluids designed for high-pressure, high-temperature (HPHT) environments. Drawing on data from 22 peer-reviewed studies published over the last decade, the research systematically evaluates the performance of key nanoparticles—including SiO_2, Al_2O_3, TiO_2, and graphene oxide (GO)—across five critical metrics: rheology, HPHT fluid loss, thermal stability, lubricity, and shale inhibition.

A weighted scorecard method was formulated to standardize and prioritize performance results with statistical methods including z-score standardization, comparison of means, and coefficient of variation analysis. Findings indicate that all nanoparticles provided performance improvements during HPHT conditions, though with differing levels of effectiveness. SiO_2 exhibited remarkable fluid loss reduction and thermal stability; GO provided better shale inhibition and lubricity; Al_2O_3 consistently improved rheological values; and TiO_2 provided robust thermal resistance but variable filtration control. Hybrid nanoparticle compositions like graphene oxide–boron nitride–titanium nitride (GO–BN–TiN) and silicon dioxide–titanium dioxide (SiO_2–TiO_2) provided synergistic benefits, leading to better overall performance compared to separate nanoparticles.

It provides a complete benchmarking of nanoparticle additives and provides quantitative insight into selecting ideal systems in HPHT drilling applications. The research focuses on hastening standardization tests and further field-scale demonstrations to bridge the gap from promise within laboratories to operational implementation. The new methodology can help with next-generation smart fluid developments compatible with extreme drilling applications.

Introduction

With the depilation of recoverable hydrocarbon reserves, the petroleum industry has been shifting toward those found in high-pressure, high-temperature (HPHT) reservoirs (Ismail et al., 2025), which are located in deepwater regions, ultra-deep formations, and re-entry wells. These HPHT wells—typically defined by bottom-hole temperatures above 150 °C and pressures exceeding 70 MPa—pose a distinct set of technical challenges (Abdelrahman et al., 2022; Ahmed et al., 2025). While they open up access to significant untapped oil and gas reserves, the extreme conditions demand sophisticated technologies throughout drilling operations, especially in the design and functionality of drilling fluids—an importance further underscored by well-control analyses showing that circulation strategy and mud rheology co-determine transient pressure responses in HPHT scenarios (Ahmed et al., 2025; Chikwe et al., n.d.; Gokapai et al., 2024a; Mehanna et al., 2024; Sajjadian et al., 2020; Sayed, 2025a).

Drilling fluid, or "mud systems," are critical to success and safety during well construction. Their primary responsibilities are to stabilize wellbore pressure, assist bearing up the borehole wall, carry cuttings to the surface, cool and lubricate throughout the bottom hole assembly, and provide a low-permeability filter cake to inhibit fluid invasion into surrounding formations. With HPHT conditions, however, conventional water-based muds (WBMs) invariably fail by degradation of polymer, elevated fluid loss, and fluid instabilities. With improved stability with oil-based muds (OBMs) and synthetic-based muds (SBMs), they are very costly, environmentally harmful, and require high-tech disposing processes (Abdullah et al., 2024; Bayat et al., 2018).

Along with rising complexity during HPHT drilling, new fluid additives to combat these challenges are gaining more and more interest. The most encouraging new technology has been nanoparticle-enhanced drilling fluids. Nanoparticles (NPs)—particles with a dimension minimum of 1–100 nm—are materials with unique properties owing to their high volume-to-surface area, tailorable-surface chemistry, and nano-scale nature. All these make them especially ideal to be deployed to enhance thermal stability, flow characteristics, and filtration potential of drilling fluids (Bayat et al., 2018; Cheraghian, 2021; Ibrahim et al., 2024).

Nanoparticles improve drilling fluid characteristics by several mechanisms:

- **Sealing Microfractures and Pores:** Due to their small size, they are capable of entering microfractures and nanopores in a rock. There they help to form a hard, impermeable film, reducing filtrate loss and improving wellbore stability.
- **Thermal Stability Reinforcement:** Thermally stable nanoparticles, including TiO_2, ZnO, and Al_2O_3, are stable to elevated temperatures and are effective to maintain fluid polymer structure, minimizing degradation with rising temperatures (Bayat & Shams, 2019).
- **Rheological Enhancement:** Nanoparticles from graphene and their variants interact with clays and polymers to stabilize or even improve fluid viscosity, yield strength, and gel properties with rising temperatures (Al-saba et al., 2018).
- **Enhanced Lubricity:** Carbon dots and graphene oxide are examples of nanomaterials providing ultra-thin lubricating films within the drill string, reducing torque and drag, primarily in horizontal or deviated wells (Su & Pan, 2025).
- **Shale Stabilization:** Functionalized nanoparticles stick to shale surfaces, inhibiting hydration, swelling, and break-up, conditions generally encountered with WBMs (Hassanzadeh et al., 2021).
- **Colloidal Stability:** The stabilized nanoparticle should not lump and stays uniformly distributed, giving a stable output during drilling.

Different nanoparticulate types were researched to satisfy these tasks, such as SiO_2, Al_2O_3, TiO_2, ZnO, GO (graphene oxide), CNTs (carbon nanotubes), carbon dots, graphitic carbon nitride (g-C_3N_4), BN (boron nitride), and hybrid materials. Performance depends on shape, chemistry, methods of dispersion, and fluid-based compatibility (Ahmed et al., 2025; Arain & Ridha, 2025; Ospanov & Kudaikulova, 2024; Su & Pan, 2025).

It gives a quantitative, systematic analysis of 20 recent papers examining nanoparticulate behavior in HPHT drilling fluids. Unlike general reviews, it uses a weighted scoring system with a focus on three key criteria: rheology, filtration control, and thermal stability.

It seeks to compare between several types of nanoparticulates, examine synergies between hybrid formulations, and help informed additive selections based on specific well and fluid conditions. The study includes tabular compilation of performance and visual scoring chart within 22 nanoparticle-containing formulations. The data are interpreted mechanistically to outline nanoparticle interaction within fluid and formation. Overall, the study hopes to add to scientists and engineers developing drilling fluids that are more efficient, safer, and environmentally friendly within HPHT conditions (Chikwe et al., n.d.).

Mechanisms of Nanoparticles in Drilling Fluids

To better understand their role in HPHT drilling environments, the following section outlines the primary mechanisms by which nanoparticles enhance fluid performance.

Sealing Nanopores and Enhancing Filtration Control

Since they are of nanoscale dimensions, particles such as Al_2O_3 as well as SiO_2 are capable of filling and sealing nanopores and microfractures present in shale and sandstone. Once they get into these features, nanoparticles establish a stable particle-size distribution, which compacts into thermostable filter cakes. This minimizes fluid loss as well as improves wellbore stability, particularly where formations are fractured, depleted, or unstable. Laboratory work suggests that HPHT filtrate loss may be reduced by up to 78.12% with nano-silica, mainly where loaded with polyethylene glycol (PEG). Hybrid nanofluids containing core materials such as $CaCO_3$, further stabilizing with CTAC surfactants, are also capable of reducing fluid loss by effectively sealing fluid paths.

Thermal Stability Reinforcement

Traditional polymers break down with increased temperature, resulting in loss of gel formation and viscosity. Nanoparticles, including TiO_2 and ZnO–TiO_2, are introduced as temperature stable reinforcement to help sustain drilling fluid polymeric network stability. The thermal stability helps with maintaining rheological characteristics and filter cake stability above 200 °C.

Rheological Property Enhancement

Nanoparticles affect viscosity, yield point (YP), and gel strength by their action with polymer chains, and with clay platelets. Chemicals, including graphene oxide (GO) and graphitic carbon nitride (g-C_3N_4), are capable of forming a three-dimensional network by van der Waals force or hydrogen bonding and thus increasing carrying capacity and drilling fluid suspension stability (Ahmed et al., 2025; Al-saba et al., 2018).

Lubricity Improvement

The use of nanoparticles like GO and carbon dots boosts lubricity by providing thin, low-friction films between tool-borehole wall contacts. This has applications where torque and drag are to be minimized, particularly in horizontal or deviated wells.

Shale Surface Modification and Inhibition

Water-based muds react negatively with reactive shale formations. Shale's surface energy is modified by nanoparticles, reducing swelling and dispersion by adsorption. Functionalized nanoparticles (e.g., Polyethyleneimine-functionalized Graphene Oxide (PEI-GO)) inhibit water invasion and contribute to enhancing wellbore stability (Abdullah et al., 2024).

Colloidal and Dispersion Stability

Nanoparticles stabilized with surfactant or polymer graft are agglomeration-free and are suspended in the fluid. The stable dispersions provide uniform performance, and sedimentation hazard is reduced during long circulation with HPHT conditions.

This combined insight into nanoparticle mechanisms, including their key contribution to microfracture sealing and filtrate invasion reduction, suggests their potential contribution to forming robust drilling fluids feasible to deploy in HPHT conditions. All these mechanisms are interrelated—one gives better filtration, which assists rheology and wellbore stability, collectively giving rationale behind better drilling fluid performance where conditions are extreme (Abdelgawad et al., 2024).

Methodology

To undertake the comprehensive evaluation of the performance of nanoparticle (NP) additives in high-performance drill fluids in the high-pressure, high-temperature (HPHT) environment, two complementary analysis strategies were employed. The first was the quantitative statistical evaluation of fluid loss data from laboratory experiments reported in the literature. This is the direct approach to gaining insight into the sealing/filtration performance of specific nanoparticles. The second was the application of a weighted scorecard model for comparatively ranking the nanoparticles against their reported impacts at the rheology of the drill-bit, filtration, and thermal stability at the drill-bit. These two analyses allowed for the combined benchmarking of NP performance against crucial operating parameters.

Statistical Evaluation Based on Fluid Loss

A focused statistical evaluation was conducted to compare the sealing performance of different nanoparticle (NP) additives in water-based drilling fluids under high-pressure, high-temperature (HPHT) conditions. The analysis centered on fluid loss behavior as a key performance indicator, using data compiled from peer-reviewed research studies that investigated a variety of nanoparticles, including SiO_2, Al_2O_3, TiO_2, graphene oxide (GO), ZnO, g-C_3N_4, boron nitride (BN), carbon nanotubes (CNTs), and hybrid systems such as SiO_2–TiO_2 and GO–TiO_2 (Arain & Ridha, 2025; Bayat & Shams, 2019; Chikwe et al., n.d.; Ospanov & Kudaikulova, 2024).

In order to allow for methodological equivalence and comparability, only experiments following standard testing conditions were considered. That is, all shortlisted experiments were performed at 0.5 wt% nanoparticle loading, 200 °C, and 500 psi pressure. All experiments also made use of the API RP 13B-1 standard or an equivalent HPHT filter press process, using water-based bentonite drilling muds. Fluid loss values of each nanoparticle composition under these constant conditions were therefore collated, and mean value, standard deviation computed from Equation 1, thereby facilitating an effective and statistically significant comparison of seal efficiency.

$$\sigma = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(X_i-\mu)^2} \quad \text{(Eq.1)}$$

Where:

- σ = standard deviation
- N = number of observations
- X_i = individual fluid loss measurement
- μ = mean fluid loss

For the quantitative study, various fluid loss values (in mL) per nanoparticle were summed up and were subjected to statistical treatment. Mean is the central tendency of the performance, while the standard deviation (STD) depicts variability as well as consistency of various studies as well as formulations.

A box plot (Figure 1) was created to graphically display the spread, the quartiles, as well as the outliers of the fluid loss behavior of each nanoparticle. Important findings are as follows:

- $GO-TiO_2$ was the best-performing NP system, having the minimum average fluid loss (9.75 mL) along with the minimum STD (0.75 mL) in the analysis, showing the best performance as well as highest reproducibility across the formulations.
- Graphene oxide (GO) also did exceptionally well, with a mean fluid loss of 7.47 mL and a standard deviation of 2.51 mL, showing excellent filtration control with moderate variability.
- $g-C_3N_4$, the carbon-based nanomaterial, recorded 8.18 mL in average, outperforming the performance of most metal oxide nanomaterials, further verifying it as the potential sealing.
- Al_2O_3 showed steady performance (STD = 1.81 mL) with the average fluid loss of 10.19 mL, indicating the system's reliability in CNTs, despite their comparatively high mean of 14.5 mL, demonstrated the lowest of all the studied material's STDs (0.5 mL) and, along with it, surprisingly steady but suboptimal fluid loss reducing performance.
- TiO_2 and ZnO were the least efficient, with high means (14.9 mL and 13.35 mL, respectively) and large standard deviations, indicative of poor as well as irregular fluid loss control (Bayat & Shams, 2019).

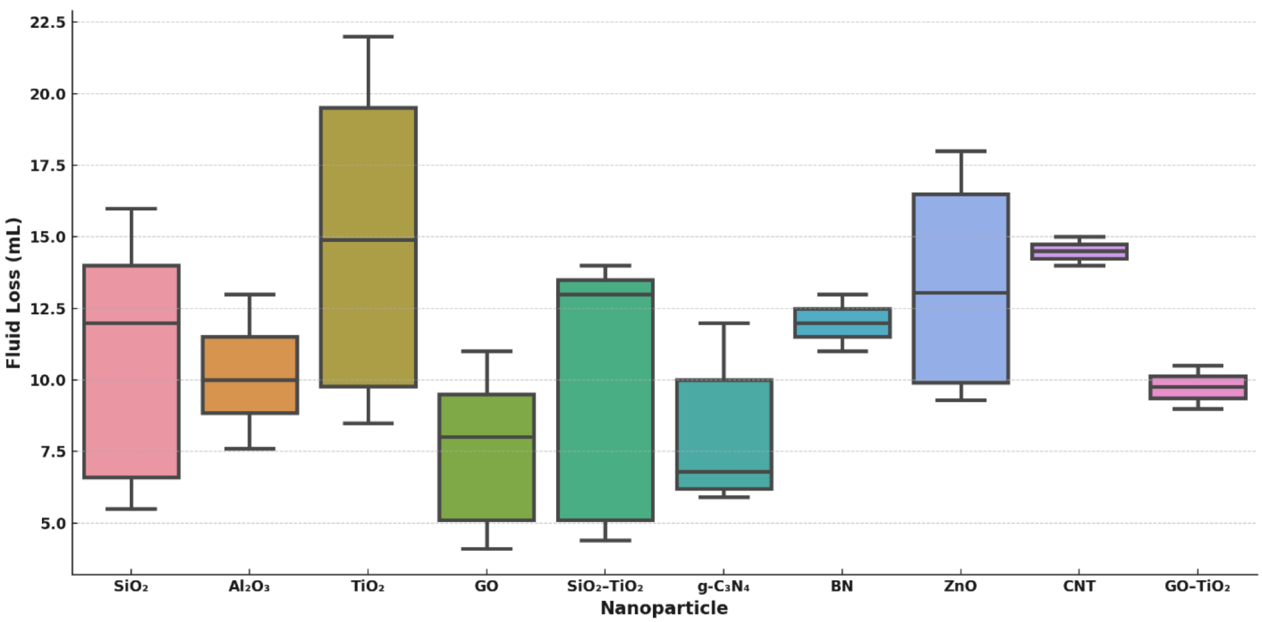

Figure 1—Box plot illustrates the fluid loss distribution (in mL) for different nanoparticle additives. The plot displays median, interquartile ranges, and outliers, providing visual insight into the variability and central tendency of each nanoparticle's sealing efficiency under HPHT conditions.

The table here below provides the summary of the statistical results and the comparative efficiency ranking by combined evaluation of the values of the mean and the standard deviation:

Table 1—Statistical summary of fluid loss performance for various nanoparticles used in water-based drilling fluids under HPHT conditions. The table presents the mean fluid loss, standard deviation (STD), and efficiency rank based on combined evaluation of performance and consistency.

Nanoparticle	Mean Fluid Loss (mL)	STD (mL)	Efficiency Rank
SiO_2	10.6	3.97	8
Al_2O_3	10.19	1.81	4
TiO_2	14.9	5.33	10
GO	7.47	2.51	2
SiO_2-TiO_2	10	4.3	7
$g-C_3N_4$	8.18	2.41	3

Nanoparticle	Mean Fluid Loss (mL)	STD (mL)	Efficiency Rank
BN	12	1	5
ZnO	13.35	3.73	9
CNT	14.5	0.5	6
GO-TiO_2	9.75	0.75	1

This assessment shows that carbon-based nanostructures and hybrid nanostructures universally surpass the classical metal oxide in reducing fluid loss at the most rigorous conditions. Additionally, the simultaneous application of the mean as one ranking parameter with the STD as another allows for better appreciation of not only peak performance but also reproducibility and stability, parameters of crucial importance for the field application in drilling operation.

Weighted Scorecard Analysis of Performance Parameters

Apart from fluid loss performance, the paper further carried out a multi-criterion analysis to estimate the wide-ranging effect of nanoparticle (NP) additives on essential performance parameters of the drilling fluid — specifically, rheological modification, filtration control, as well as thermal stability. This was done using the weighted scorecard method, integrating the outcomes of experiments reported in various peer journal articles.

Each nanoparticle was rated in three key performance areas on a qualitative scale:

- Rheological Enhancement: Enhancement of viscosity, yield point (YP), as well as gel strength behavior (Al-saba et al., 2018).
- Filtration Control: Reductions in API/HPHT fluid loss and formation of high-quality filter cakes.
- Thermal Resistance: Capability of sustaining functional performance and stability at increased temperature conditions.

For every domain, the NPs were given qualitative scores — "No improvement", "Moderate improvement", or "Significant improvement" — which were calibrated to the numerical score in the 0–3 range, with scores for each domain being weighted and averaged. Domain weights were set at 40% filtration, 30% rheology, and 30% thermal stability — which reflects the high priority of filtration performance for HPHT conditions.

Results were visualized in Figure 2, presenting average performance scores for different NPs:

- Green bars (score ≥ 2.0) suggest high-performing NPs like ZrO_2 (1 wt%), Nano-SiO_2 (grafted), and Al_2O_3 (α & γ) — that had always outperformed others in the three measurements (Medhi et al., 2020; Pang et al., 2025).
- Yellow bars (score ≈ 1.2–1.9) indicate the NPs of medium effectiveness, i.e., SiO_2-TiO_2, Silica-polymer, and Hybrid oxide, which provided improvement but whose effectiveness is system design- and condition-dependent(Prakash et al., 2021).
- Red bars (score < 1.2) include NPs like Carbon dots, PEI-GO, and SiO_2 (varied sizes), which showed limited or inconsistent enhancements, especially under HPHT scenarios.

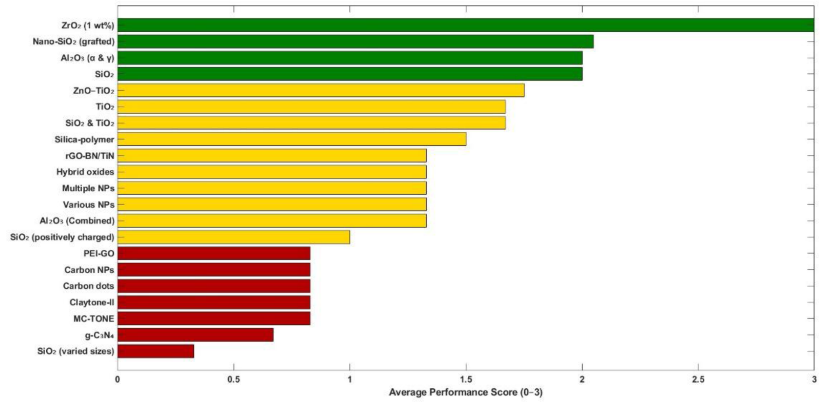

Figure 2—Color-coded efficiency of nanoparticles in HPHT drilling fluids based on rheology, filtration, and thermal stability.

Table 2 summarizes the particular qualitative evaluation for each NP in the three categories with essential references for these evaluations. For example, the performance of ZnO–TiO_2 was found to present "significant" improvement in rheology as well as in filtration, with "excellent" thermal stability, while for SiO_2 (positive charged) excellent filtration benefits were observed but inadequate rheological and thermal studies were provided (Ibrahim et al., 2024) .

Table 2—Performance summary of nanoparticle additives in HPHT drilling fluids based on rheological enhancement, filtration control, and thermal stability metrics.

Nanoparticle	Rheology	Filtration	Thermal Stability	Reference
Nano-SiO_2 (grafted)	Improved	Excellent	Very good	(Seidina Ousseini et al., 2025)
SiO_2 (charged)	Not detailed	Improved	Not mentioned	(Giraldo et al., 2017)
SiO_2	Improved	78.12%	Enhanced	(Prakash et al., 2021)
Al_2O_3	Improved	Improved	Improved	(Anoop et al., 2016a; Smith et al., 2018)
Al_2O_3 (α & γ)	Improved	60%	Confirmed	(Hassanzadeh et al., 2021)
Al_2O_3 in silicone oil	Increased	Not discussed	Pressure stable	(Anoop et al., 2016b)
TiO_2	Improved	18–28% ↓	Improved	(Beg et al., 2020)
SiO_2 & TiO_2	Enhanced	1.0 wt% best	Improved	(Kumar et al., 2020)
rGO-BN/TiN	Up to 89% YP ↑	Strong ↓	230°C	(Sayed, 2025b)
Various NPs	Mentioned	Strong ↓	Enhanced	(Cheraghian, 2021)
rGO-BN/TiN (alt)	Marked	Reduced	Strong	(Sayed, 2025a)
g-C_3N_4	YP ↑ 66%	68% ↓	Improved	(Ahmed et al., 2025)
PEI-GO	Significant	67% ↓	160°C	(Abdullah et al., 2024)
ZnO–TiO_2	Significant	Strong	Excellent	(Bayat & Shams, 2019)
Carbon NPs	Yes	Improved	Strong	(Lysakova et al., 2023)
Hybrid oxides	Moderate to high	Strong	Enhanced	(Cheraghian, 2021; Gokapai et al., 2024b)
Silica-polymer	Improved	High	Good	(Cheng et al., 2025)
Carbon dots	Moderate	Very strong	Good	(Zhong et al., 2025)
Claytone-II	High G′, G″; Strong gel	4.6 mL	Implied good	(Mahmoud et al., 2024a)
MC-TONE	High YP/PV; Good gel	5.0 mL	Implied good	(Mahmoud et al., 2024b)
ZrO_2 (1 wt%)	Significant viscosity/ elasticity increase	Minimum filtrate loss	Stable up to 80 °C	(Medhi et al., 2020; Sajjadian et al., 2020)

This scorecard analysis offers a complementary tool to the fluid loss-based ranking system, whereby NPs can be selected not only on the basis of sealing effectiveness but also based upon their capacity to maintain broader drilling fluid performance in the presence of operational stressors.

Nanoparticle Characterization

The functional performance of nanoparticle -based drilling fluids during HPHT environments closely relates to nanoparticles' physicochemical properties. Even though the present meta-analysis emphasizes functional performance indicators, evidence from literature references indicates important trends of nanoparticle morphology, nanoparticle size, nanoparticle dispersion, and nanoparticle surface chemistry, impacting drilling fluid response (M. Ali et al., 2020).

Particle Size and Morphology

Most effective nanoparticles, including SiO_2, Al_2O_3, and TiO_2, were reported with average particle sizes ranging from 10 to 100 nm. Smaller particles demonstrated superior sealing of nanopores and contributed to more compact filter cakes. For example, silica nanoparticles with an average diameter of 20 nm significantly reduced HPHT fluid loss by enhancing the sealing of mud cakes. Graphene oxide and carbon-based nanomaterials exhibited sheet-like or layered morphologies, providing large surface areas and interfacial contact. This structure contributes to the observed improvements in rheology and shale inhibition, as reported in multiple studies (Alkalbani et al., 2023; Prakash et al., 2021).

Surface Area and Functionalization

Nanoparticles with higher specific surface area exhibited improved adsorption onto clay surfaces and improved fluid component interactions. Functionalization (e.g., hydroxyl, amine, or carboxyl groups) improved base fluid compatibility as well as thermal cycling stability. In particular, graphene oxide gained an advantage from oxygenated functional groups providing hydrogen bonding and dispersion within WBMs. Analogously, surface-functionalized Al_2O_3 particles improved colloidal stability and rheological synergy with polymer applications (Smith et al., 2018).

Dispersion Stability

Robust dispersion proves important to avoid agglomeration of particles, which decreases surface area and counteracts positive effects. Some research highlighted surfactants or polymeric stabilizers to achieve stable nanoparticle distribution even during thermal stress. Inconsistent performance resulted from poor dispersion, as sometimes encountered with TiO_2 (Beg et al., 2020).

Thermal and Chemical Stability

Nanoparticles were also required to maintain structural integrity during HPHT conditions. Compositions including TiO_2 and Al_2O_3 showed remarkable thermal stability, with retention of crystallinity and function beyond 200 °C. Chemical inertness was also a consideration, particularly for long-period exposure to hostile downhole conditions.

To effectively assess these physicochemical characteristics, several characterization methods have been utilized among the studied literature. Some of these involve scanning and transmission electron microscopy to analyze morphology, XRD to evaluate crystallinity, FTIR to identify surface functional moieties, and zeta potential measurements to assess colloidal stability, among others. A compilation of the most employed characterization tools along with their application to drilling fluid characteristics is tabulated below in Table 3.

Table 3—Characterization techniques used in nanoparticle-enhanced drilling fluid studies and their relevance to performance assessment under HPHT conditions.

Instrument / Technique	Purpose / Information Provided	Importance in Drilling Fluid Context
SEM (Scanning Electron Microscopy)	Surface morphology and particle size visualization	Confirms nano-scale structure and surface topology, affecting dispersion and sealing ability.
TEM (Transmission Electron Microscopy)	Internal structure and detailed nano-architecture	Used to verify crystal shape and size for advanced engineered nanomaterials.
XRD (X-ray Diffraction)	Crystalline structure and phase identification	Determines if nanoparticles are amorphous or crystalline, influencing thermal stability.
FTIR (Fourier Transform Infrared Spectroscopy)	Surface functional groups and chemical bonding	Confirms surface modifications (e.g., grafting, capping), important for wettability and reactivity.
Zeta Potential Analyzer	Surface charge and colloidal stability	Indicates dispersion quality and repulsion forces, critical for long-term stability in fluid systems.
DLS (Dynamic Light Scattering)	Particle size distribution in suspension.	Ensures uniform dispersion and size control, directly affecting rheological behavior.
TGA (Thermogravimetric Analysis)	Thermal degradation profile.	Assesses thermal resilience of modified particles under HPHT conditions.
BET Surface Area Analysis	Specific surface area and porosity	High surface area increases reactivity and improves performance in filtration control.
UV–Vis Spectroscopy	Optical properties and dispersion confirmation	Supports surface modification confirmation and dispersion behavior (e.g., GO and TiO_2).

Results and Discussion

In this research, we used two complementary techniques—a statistical evaluation of fluid loss data and a weighted scorecard system—in evaluating and comparing the performance of selected nanoparticle additives used in water-based drilling fluids (WBMs) under HPHT conditions. Statistical evaluation allowed quantification of consistency and sealing efficiency from one study to another, while the weighted scorecard offered a qualitative but systematized comparison of rheological improvement, thermal stability, and fluid loss reduction. Both, together, provide an overall performance rank of the nanoparticles under conditions feasible for operations. Among all of the research that were discussed, silicon dioxide (SiO_2) as well as graphene oxide (GO) always distinguished themselves from the rest of the additivities. GO had notable multifunctionality, as numerous studies revealed improved shale inhibition, lubricity, as well as rheological stability. Specifically, GO-based systems had structural stability when subject to thermal aging at 230 °C as well as standard increased gel strength as well as shear stress measurements. GO also achieved 78.12% fluid loss lowering, which has been associated with GO's enormous surface area as well as sheet-like structure that facilitated stable filter cake formation as well as phase dispersion of the fluid (Ahmed et al., 2025).

Silicon dioxide (SiO_2) enjoyed exceptional filtration control, achieving filtrate volume reductions of 75% or more, especially when functionalized or as nanoscale dispersion systems. High surface area and fine particulate size offered compact cake development and substantial thermal degradation resistance, especially at >200 °C temperatures and >500 psi pressures. But performance varied as a function of differing formulations, which suggested sensitivity to fluid phase as well as nano dispersion quality. Aluminum oxide (Al_2O_3) was mainly effective at improving rheological performance. In one significant study, plastic viscosity and yield point retention improved following treatment at 220 °C, yielding corresponding HPHT filtrate loss reductions of over 60%. Although there were such benefits, filtration performance of Al_2O_3 was less predictable, implying a degree of dependence on fluid treatment formulation and inter-particle interaction.

Thermal and rheological stability of titanium dioxide (TiO_2) were average. Some tests demonstrated good viscosity retention at elevated temperatures (e.g., 180 °C), but fluid loss performance was variable. The variability most likely results from tendencies for agglomeration, which dampen surface activity as well as dispersion stability, which restrict TiO_2 performance in dynamic systems. The work also points to the advantage of hybrid nanomaterials over the single-phase nanomaterials. Combinations of complementary nanomaterials resulted in better performance across several domains. For example:

- The GO–BN–TiN nanocomposites displayed enhanced thermal tolerance and enhanced rheology after 230 °C, which surpassed their building blocks (Sayed, 2025).
- SiO_2–TiO_2 blend enhanced fluid loss control along with preservation of gel strength, which offered synergies when exposed to thermal stress.
- The GO-g-C_3N_4 system showed exceptional performance of thermal stability of WBMs and filtration efficiency of pre-aged WBMs at 200 °C, which resulted from improved interfacial effects as well as enhanced hydrogen bonding.

These findings highlight the promise of rationally designed hybrid nanoparticles to overcome several HPHT issues at one time. Here, we deliver the mean of fluid loss, standard deviation, as well as efficiency ranking from the statistics analysis, offering a precise quantitative reference value of seal performance. At the same time, Figure 1 displays the weighted average score from the scorecard system so that comparative studies of rheological, thermal, as well as filtration, sections are attainable. From a practical perspective:

- They are suitable for applications where low fluid loss and thermal stability are of prime importance for WBMs.
- Al_2O_3 finds use where greater rheological control must be achieved, as when deep vertical well or sag-sensitive formations and settling of cuttings are of concern.
- Hybrid systems are best suited to complicated HPHT environments where performance from several domains is needed.

However, although such initial findings hold promise, variability of performance remains a recurring issue. Outcome strongly depends on nanoparticle concentration, base fluid chemistry, particle distribution, as well as on test procedures. Nonuniformity of methods used among studies prevents direct comparison as well as generalization of results.

In addition, although there are various references to shale inhibition and lubricity, data inconsistencies, as well as quantitative reporting, prevented them from becoming part of the scorecard model. These are, however, relevant properties of HPHT extended-reach and deviated wells, and as such, future work should incorporate standardized measurements as well as quantitative results for such properties. Overall, this hybrid methodology framework ranks existing nanoparticle options as well as provides strategic guidance for HPHT drilling fluid system optimization to fluid engineers as well as designers.

Temperature Sensitivity of Yield Point for Nanoparticle-Based Drilling Fluids

Rheological properties, particularly yield point (YP), of drilling fluids are a critical controlling parameter of suspension capacity and carrying capacity of drilling fluids. The properties could be influenced by variations in temperature, particularly those faced with high-pressure high-temperature (HPHT) drilling. Nanoparticles are researched and used broadly as admixtures to enhance thermal stability and flow resistance of drilling fluids. To evaluate temperature's influence on yield point behavior across several nanoparticle suspensions, the comparative research revealed apparent thermal performance trends across nanoparticle types. Upon an increase from 25 °C to 100 °C, suspensions all showed a decrease in yield point, but levels of degradation ranged substantially.

As illustrated from Figure 3, multi-walled carbon nanotubes (MWCNT) and titanium dioxide (TiO_2) exhibited greater yield point values across the full temperature range, indicating improved thermal resistance. In contrast, silicon dioxide (SiO_2) exhibited a more pronounced shift, indicating less structural retention with rising temperature. Zinc oxide (ZnO) and boron nitride (BN) showed reasonable performance with predominantly linear degradation trends (Arain & Ridha, 2025). The findings highlight nanoparticle selection's importance during design considerations among thermally stable HPHT drilling fluids.

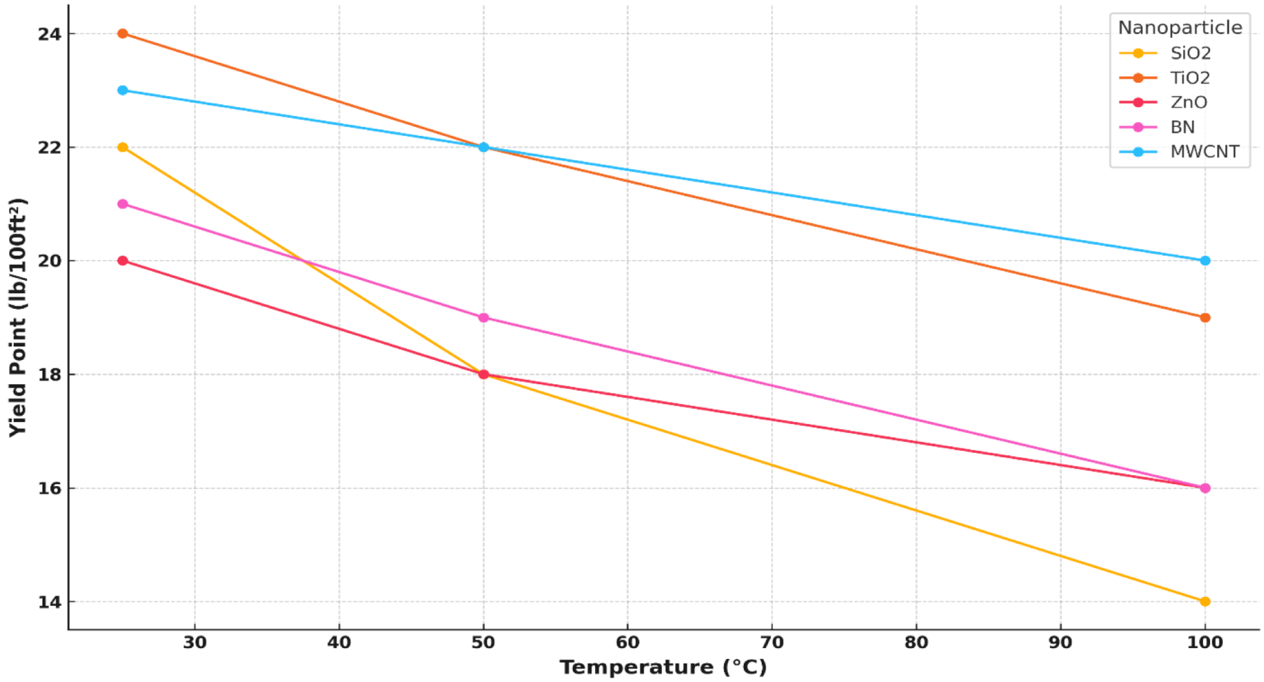

Figure 3—Effect of temperature on the yield point of nanoparticle-enhanced drilling fluids. Yield point values were extracted from multiple sources and plotted across a range of 25 °C to 100 °C. MWCNT and TiO_2-based fluids exhibited better thermal retention compared to SiO_2, BN, and ZnO systems. Each data point represents average reported values from corresponding literature.

Effect of Nanoparticles on Fluid Shear Response

Figure 4 represents rheological properties of nanoparticle-enhanced drilling fluids with variable thermal and pressure conditions, compared to reference fluids, functionalized nanofluids, and drilling fluids with high-temperature and high-pressure (HPHT) conditions. The shear stress-shear rate relationship represents a primary indicator of a fluid's ability to suspend cuttings, achieve wellbore stability, and lubricate drilling equipment.

The data unmistakably shows that nanoparticles significantly augment drilling fluid shear stress across the full shear rate range. For instance, 0.5 wt% silica nanoparticle fluid consistently exhibits higher shear stress than the baseline fluid, especially with higher shear rates. This improvement primarily comes from a large nanoparticle surface area and interaction capability, aiding fluid structure and internal resistance during dynamic flow (Kök & Bal, 2019).

Similarly, PEI-GO (graphene oxide functionalized with polyethyleneimine) at 0.3 wt% exhibits a remarkable increase in shear stress with respect to the base fluid. This could be a result of PEI-GO functional groups enhancing interparticle interaction and increasing dispersion stability, leading to better load transmission and structural strength during shear. Also, during 100 °C tests, the high-temperature base fluid shows higher shear stress with increased nanoparticle concentration, indicating nanoparticle-modified fluids are resistant to thermal stresses with persistent performance. However, even though there is a loss in shear stress, the fluid tested with HPHT conditions (200 °C and 14,000 psia) shows higher shear stress still, as compared to an unmodified base fluid tested with similar conditions. This indicates thermal sensibility of drilling fluids and partial robustness met with nanoparticle additions (Abdullah et al., 2024; Su & Pan, 2025).

Typically, nanoparticle addition increases drilling fluid's viscoelastic characteristics, providing better cutting transportation, reduced sag, and improved wellbore cleanup. This advantage becomes particularly important in HPHT wells, where regular fluids might be negatively impacted by degradation or loss of rheological characteristics. Better thermal and pressure stability by nanoparticle-laden fluids result in a more productive, safer drilling operation, with a reduction in non-productive time, and a decrease in risks from wellbore instability and fluid loss.

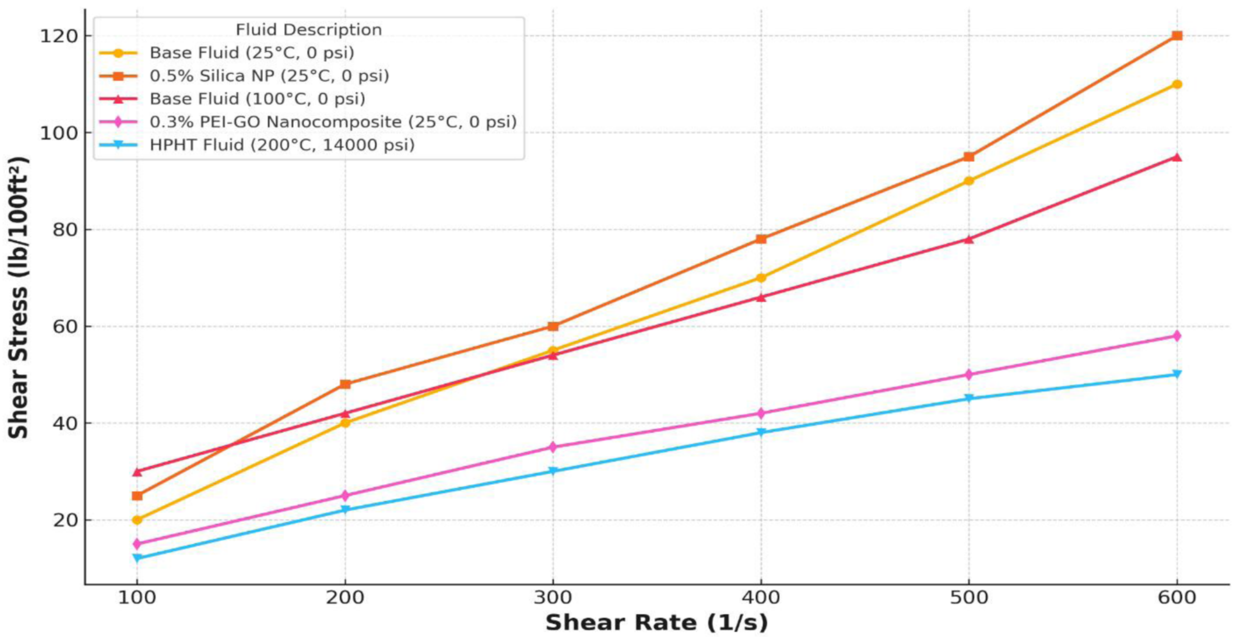

Figure 4—Shear stress–shear rate profiles for various nanoparticle-enhanced and base drilling fluids. The figure compares the rheological behavior of a base fluid at room conditions, base fluid with 0.5 wt% silica nanoparticles, 0.3 wt% PEI-GO nanocomposite, and the same fluid under elevated temperature (100 °C) and HPHT (200 °C, 14,000 psi) conditions. The enhancement in shear stress from nanoparticle addition demonstrates improved structural stability and potential drilling efficiency under dynamic flow and extreme environments.

Limitations and Challenges

Even though HPHT drilling fluid applications have been optimistically enhanced by nanoparticles, several limitation factors exist, though, which suppress their wide-scope industrial application as shown in table 4.

Table 4—Key Challenges in the Application of Nanoparticles in Drilling Fluids Under HPHT Conditions.

Challenge Area	Description
Dispersion and Stability	Nanoparticles tend to agglomerate, especially under high salinity and temperature. Surfactants or functionalization may be needed, increasing complexity and cost.
Scale-Up and Field Validation	Most research is limited to lab-scale formulations. Few field trials exist, making it difficult to scale up results to operational environments.
Economic Viability	High-purity or functionalized nanoparticles (e.g., graphene oxide) are costly, which may hinder their use in cost-sensitive drilling applications (Su & Pan, 2025).
Environmental and Regulatory Concerns	The long-term environmental impact of nanoparticles in subsurface conditions is unclear. Regulation of nano-enhanced fluids is still developing.
Lack of Standardization	There is no standardized testing protocol for nanoparticle performance under HPHT conditions, limiting comparability and benchmarking.

Future Work and Research Directions

The integration of nanoparticles into drilling fluids holds immense promise for improving performance under high-pressure, high-temperature (HPHT) conditions. However, to transition from laboratory research to widespread field adoption, targeted advancements are needed. Table 5 outlines key future research directions aimed at overcoming current limitations, enhancing functionality, and ensuring the safe and effective deployment of nano-enhanced fluids in real-world drilling operations.

Table 5—Future Research Directions for Advancing Nanoparticle-Enhanced Drilling Fluids in HPHT Applications

Future Research Direction	Description
Field Testing and Validation	More pilot-scale and full-field deployments are needed to validate laboratory findings under actual drilling conditions, including deepwater and ultra-HPHT environments.
Advanced Surface Engineering	Development of thermally stable and chemically compatible nanoparticle coatings or grafts is needed to improve long-term performance and reduce aggregation.
Smart and Responsive Nanoparticles	Stimuli-responsive nanoparticles (e.g., temperature-activated, pH-sensitive) could enable smart fluids that adapt to downhole environments in real-time.
Life-Cycle and Environmental Impact Assessment	Comprehensive studies on the fate, transport, and ecotoxicity of nanoparticles are essential for safe and sustainable use in drilling applications.
Standardized Evaluation Protocols	Establishing standardized HPHT testing protocols (e.g., rheology, filtration, thermal resistance) would improve comparability and industry acceptance (Ibrahim et al., 2024).

Conclusion

This research comprehensively analyzed nanoparticle-based drilling fluids by combining statistical fluid loss evaluation, rheological performance evaluation, and multi-domain performance scorecard assessments. Ten different nanoparticle systems were analyzed—SiO_2, Al_2O_3, TiO_2, graphene oxide (GO), ZnO, g-C_3N_4, carbon nanotubes (CNTs), BN, and two hybrid systems (SiO_2–TiO_2 and GO–TiO_2).

The fluid loss analysis, upon consideration of peer-review articles, involved determination of the mean as well as the standard deviation (STD) of the fluid loss values in HPHT conditions. Analysis showed GO–TiO_2 had the least fluid loss mean of 9.75 mL with extremely low STD of 0.75 mL, indicative of steady performance as well as high sealing capacity. TiO_2, on the contrary, had the highest fluid loss of 14.9 mL as well as the highest variability, indicative of less steady performance.

The studies also observed rheological properties, shear stress, yield point (YP), as well as shear rate tendencies, whereby various nanoparticles increased viscosity, as well as gel strength. In particular, g-C_3N_4 increased the size of the YP by 66%, while rGO-BN/TiN showed up to 89% of this parameter's rise with the retention of thermally induced instability.

Lastly, a qualitative scorecard method was utilized to integrate observations on filtration control, thermal resistance, and rheology from the results of experimentation in literature. This method verifies that hybrid as well as carbon-based nanomaterials, e.g., PEI-GO and ZnO–TiO_2, offer multi-domain improvements applicable in real-field drilling applications.

Overall, this multi-metric analysis further emphasizes the high promise of graphitic as well as hybrid nanoparticles for improving HPHT drilling fluids performance. This offers useful guidelines for the selections of nanoparticle additives optimally designed for thermal resistance, fluid loss, as well as rheological improvement.

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